

TOPIC 4/60

Evolution of Language Models

RNN → LSTM → Transformer

How AI finally learned to read context.

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THE CORE CHALLENGE

Unlike image recognition, language is **sequential**. Words only make sense based on the words that came before them.

"The cat, which already ate a huge bowl of food, ___ full."

To predict "was" (instead of "were"), the AI has to remember the word "cat" from the beginning of the sentence.

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ERA 1: RNNS



Recurrent Neural Networks

The early solution. RNNs process text one word at a time, left-to-right, maintaining a "loop" of information.

How it works: It reads word #1, updates its hidden state, passes that state to word #2, and repeats. Like passing a baton in a relay race.

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THE FATAL FLAW

Short-Term Memory

Vanishing Gradients

RNNs are terrible at long sentences. As the context gets passed down the chain, the earliest words slowly fade away.

The Problem:

By the time an RNN reaches the end of a paragraph, it has completely forgotten what the first sentence was about.

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ERA 2: THE FIX

LSTM

Long Short-Term Memory

Engineers added a "conveyor belt" (cell state) and special "gates" to the RNN.

The Gates: They act like bouncers, explicitly deciding: "Keep this word in memory" or "Forget this word, it's irrelevant." This fixed the short-term memory issue!

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THE NEXT BOTTLENECK

The Speed Limit

Sequential Processing

LSTMs were smart, but they still had to read sequentially (Word 1 → Word 2 → Word 3).

Why it failed scaling:

You couldn't train them using parallel processing on thousands of GPUs. You had to wait for word 1 to finish before processing word 2.

> Training on the whole internet would take centuries.

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ERA 3: THE BREAKTHROUGH

2017: GOOGLE RESEARCH

"Attention Is All You Need"

The paper that literally changed the world. It proposed throwing away the sequential RNN loop entirely.

THE RESULT

The Transformer architecture was born. (The "T" in ChatGPT).

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THE MAGIC TRICK

Self-Attention

Instead of reading left-to-right, Transformers look at all the words at the exact same time.

By looking at everything at once, it mathematically calculates which words pay "attention" to each other, forming instant, flawless context webs regardless of how long the sentence is.

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WHY TRANSFORMERS WON

Massive Parallelization

Because it doesn't read sequentially, you can split the training across 10,000 GPUs simultaneously.

Infinite Context

No more short-term memory loss. It remembers the first word as clearly as the last.

The Birth of LLMs

This specific speed breakthrough is what allowed us to train models on the *entire internet*, creating modern GenAI.

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Evolution of Models

CONCEPT UNLOCKED



RNN: Sequential loop (Forgot things)



LSTM: Added memory (Too slow)



Transformer: All at once (Changed the world)

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